

Chapter 1 The Foundation of Logic and Proofs

1.1 Propositional Logic

Why Are We Studying Logic?

Thinking logically is one of the most useful skills you can develop.

Logic is the basis of computer science.
Programs are more or less pure logic.

Logic: The Foundation of Reasoning

What is formal logic?

Foundation for organized and careful method of thinking that characterizes reasoned activity.

The study of reasoning, specifically concerned with whether something is true or false.

Note: Formal logic focuses on the relationship between statements as opposed to the content of a particular statement

Propositions

Definition: Proposition (also called Statements)

A proposition is a statement that is either true or false.

Examples:

The Arizona men's basketball team won the NCAA championship in 1997

True

Dr Anson played on that team.

False

3.14 is an irrational number

False

π is an irrational number

True

Propositions

The following are not propositions:

Whatcha doing?

Question, not true or false so not a proposition

Is your name Ferb?

Even a yes/no question is not a proposition.

$x^2 = 4$

What is x? The truth depends on the status of the variable.

Everybody dance now.

Command, not a proposition.

Propositions

Which of the following are propositions?

- Everyone has seen Phineas and Ferb
- 7 is greater than -3
- Where's Perry?
- $x + y = z$
- Anyone who wears a hat is from England.

Propositions

Which of the following are propositions?

- Everyone has seen Phineas and Ferb
- 7 is greater than -3
- ~~Where's Perry?~~
- ~~$x + y = z$~~
- Anyone who wears a hat is from England.

Proposition Labels

To save writing, it is traditional to label propositions with lower-case letters

Examples:

p: The Arizona men's basketball team won the NCAA championship in 1997

q: Dr Anson played on that team.

h: 3.14 is an irrational number

p is True, q is False, h is also False

Compound Propositions

Definition: Compound Proposition

A combination of propositions which are connected by a logical operators. (Note: compound propositions are also propositions)

A proposition that isn't a compound proposition is called an **atomic or **simple** proposition.**

The first logical operator we'll talk about is conjunction.

Conjunctions

Conjunctions are:

- compound propositions formed in English with the word “and”,
- formed in logic with the symbol (“ $\wedge$ ”), and
- true only when both participating propositions are true.

Example(s):

Conjunctions

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Example(s):

p: I bought gas.

q: I got a haircut.

Conjunctions

Conjunctions are:

- compound propositions formed in English with the word “and”,
- formed in logic with the symbol (“ $\wedge$ ”), and
- true only when both participating propositions are true.

Example(s):

p: I bought gas.

q: I got a haircut.

$p \wedge q$: I bought gas and I got a haircut.

Truth Table

A truth table is often used to show the value of a compound proposition for every possible value of the simple propositions that make it up. For conjunction it looks like this:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

More Examples

Your instructor's name is Dr Anson and he is 6'10" tall.

Is this a true or false proposition?

What are the simple propositions that make it up?

Are they true or false?

More Examples

Your instructor's name is Dr Anson and he is 6'10" tall.

Is this a true or false proposition?

What are the simple propositions that make it up?

Are they true or false?

p: Your instructor's name is Dr Anson.

q: He is 6'10" tall.

Proposition: $p \wedge q$

p is true and q is false so the proposition is false.

English and Conjunctions

There are many ways in English to indicate that two propositions are true. Their meaning in English varies somewhat, but their logical meaning is the same:

p: Shaggy was afraid.

q: The kids foiled the crook.

All the following translate to $p \wedge q$:

Shaggy was afraid, but the kids foiled the crook.

Despite the fact that Shaggy was afraid, the kids foiled the crook.

Although Shaggy was afraid, the kids foiled the crook.

Disjunctions

Disjunctions are:

- propositions formed in English with the word “or”,
- formed in logic with the symbol “ $\vee$ ”, and
- true when at least 1 of the participating propositions are true.

Example:

n: Your instructors name is Dr Anson

t: He is 6’10” tall

$n \vee t$: Your instructors name is Dr Anson or he is 6’10” tall.

Proposition is true.

Disjunctions

The truth table for disjunctions looks like:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Problems with English

While in mathematics, formal logic, and computer science “or” means disjunction (the prop is true if one or both of the joined props is true), in spoken English the meaning is not so clear.

For Example: you love or hate this class

The speaker doesn't mean you can both love and hate the class. One or the other will happen. In other words, EXACTLY one must be true. We call this an exclusive or.

Exclusive or

$$p \oplus q$$

True if p is true or q is true but not both are true.

$\oplus$ is always called exclusive or

$\vee$ is sometimes called inclusive or to emphasize the difference

Negation

Negating a proposition simply flips its value.

Symbols representing negation include:

$$\neg X \quad \bar{X}$$

Example(s):

h : Homework 1 is easy

$\bar{h}$: Homework 1 is not easy

Conditional Propositions

Definition: Conditional Proposition

A proposition expressible in the form “if p , then q ” ($p \rightarrow q$, read “ p implies q ”), where p and q are propositions, is a conditional proposition.

Example:

If Sauron gets the ring, then all Middle-earth will be destroyed.

Conditional Propositions

The statement: "If p then q" is written in logic as:

$$p \rightarrow q$$

Where p is called the hypothesis and q is called the conclusion.

Here is the truth table for a conditional statement:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Conditional Propositions

There are many ways to express conditional propositions in English. Here are a few:

If p , then q	If Sauron gets the ring, then all Middle-earth will be destroyed.
If p , q	If Sauron gets the ring, all Middle-earth will be destroyed.
q if p	All Middle-earth will be destroyed if Sauron gets the ring
p implies q	Sauron getting the ring implies all Middle-earth will be destroyed.
p only if q	Sauron gets the ring only if all Middle-earth is destroyed.
p is sufficient for q	Sauron getting the ring is sufficient for all Middle-earth to be destroyed.
q is necessary for p	All Middle-earth being destroyed is necessary for Sauron to get the ring

Biconditional Propositions

What is the meaning of:

A triangle is equilateral if and only if all three angles are equal.

Biconditional Propositions

What is the meaning of:

A triangle is equilateral if and only if all three angles are equal.

t: A triangle is equilateral a: All three angles are equal

IF AND ONLY IF

Biconditional Propositions

What is the meaning of:

A triangle is equilateral if and only if all three angles are equal.

t: A triangle is equilateral a: All three angles are equal

IF AND ONLY IF

t if a

if a, then t

$(a \rightarrow t) \quad \wedge$

Biconditional Propositions

What is the meaning of:

A triangle is equilateral if and only if all three angles are equal.

t: A triangle is equilateral a: All three angles are equal

IF	AND	ONLY IF
t if a		t only if a
if a, then t		if t, then a
$(a \rightarrow t)$	$\wedge$	$(t \rightarrow a)$

Biconditional Propositions

Definition: Biconditional Proposition

The expression “p if and only if q” (“p iff q”) is called the biconditional proposition (a.k.a. biimplication) and is represented by the symbol $\leftrightarrow$ (as in: $p \leftrightarrow q$).

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r	s	$r \rightarrow s$	$s \rightarrow r$	$(r \rightarrow s) \wedge (s \rightarrow r)$	$r \leftrightarrow s$
T	T				
T	F				
F	T				
F	F				

Biconditional Propositions

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r	s	$r \rightarrow s$	$s \rightarrow r$	$(r \rightarrow s) \wedge (s \rightarrow r)$	$r \leftrightarrow s$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Evaluating Compound Propositions

Does the order we evaluate operations matter?

Consider $F \wedge F \vee T$

If we evaluate the conjunction (and) first we get

$$\underline{F \wedge F} \vee T \Rightarrow F \vee T \Rightarrow T$$

If we evaluate the disjunction (or) first we get

$$F \wedge \underline{F \vee T} \Rightarrow F \wedge T \Rightarrow F$$

So, the order does matter.

Order of operations

1. $\neg$ (not)
2. $\wedge$ (and)
3. $\vee$ (or)
4. $\rightarrow$ (if)
5. $\leftrightarrow$ (iff)

$$\text{So: } p \vee \neg q \wedge h = p \vee ((\neg q) \wedge h)$$

One way to remember this is to think of evaluating numbers:

$$x - y * z = x + ((-y) * z)$$

So, you can think of not as negative, and as multiplication, and or as division.

You can also always use parenthesis to make the order clear.

Order of operations

1. $\neg$ (not)
2. $\wedge$ (and)
3. $\vee$ (or)
4. $\rightarrow$ (if)
5. $\leftrightarrow$ (iff)

What about $\oplus$?

The order of operations is not consistently defined for other logical operators (we'll see more soon). So, anything involving any others should include parenthesis to indicate the order.

Evaluating Compound Propositions

To evaluate a compound proposition, simplify each step in order of precedence until a value of T/F is arrived at:

Ex: $\neg (p \vee q) \wedge r$ where $p = T, q = F, r = T$

$\neg (T \vee F) \wedge T$ - fill in values

$\neg T \wedge T$ - evaluate in parenthesis

$F \wedge T$ - evaluate not

F - final answer

Truth Tables

We often want to show the possible values of a proposition for all combinations of values of the simple propositions. To do this we use a truth table. Here's what a truth table looks like:

p	q	r	$\neg q$	$(p \vee r)$	$\neg q \wedge (p \vee r)$
T	T	T	F	T	F
T	T	F	F	T	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	F	F	F
F	F	T	T	T	T
F	F	F	T	F	F

Truth Table Order

For consistency there's an agreed order for the rows. First know if there are n simple propositions then there are 2^n rows. We'll do an example where $n = 3$.

p	q	r

First create a table with 8 rows

Truth Table Order

p	q	r
T		
T		
T		
T		
F		
F		
F		
F		

For the first proposition make the first half T and the second half F

Truth Table Order

p	q	r
T	T	
T	T	
T	F	
T	F	
F	T	
F	T	
F	F	
F	F	

Now alternate 2 T's with 2 F's starting with T

Truth Table Order

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Lastly alternate Ts & Fs

Example Truth Table

Let's do the truth table for $\neg (p \vee q) \wedge r$

p	q	r	$p \vee q$	$\neg (p \vee q)$	$\neg (p \vee q) \wedge r$

First create a table with 8 rows and a column for each step of the evaluation.

Example Truth Table

Let's do the truth table for $\neg (p \vee q) \wedge r$

p	q	r	$p \vee q$	$\neg (p \vee q)$	$\neg (p \vee q) \wedge r$
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

Now fill in the values for the simple propositions as described before.

Example Truth Table

Let's do the truth table for $\neg (p \vee q) \wedge r$

p	q	r	$p \vee q$	$\neg (p \vee q)$	$\neg (p \vee q) \wedge r$
T	T	T	T		
T	T	F	T		
T	F	T	T		
T	F	F	T		
F	T	T	T		
F	T	F	T		
F	F	T	F		
F	F	F	F		

Then fill in the columns one at a time using the values of the previous columns.

Example Truth Table

Let's do the truth table for $\neg (p \vee q) \wedge r$

p	q	r	$p \vee q$	$\neg (p \vee q)$	$\neg (p \vee q) \wedge r$
T	T	T	T	F	
T	T	F	T	F	
T	F	T	T	F	
T	F	F	T	F	
F	T	T	T	F	
F	T	F	T	F	
F	F	T	F	T	
F	F	F	F	T	

Example Truth Table

Let's do the truth table for $\neg (p \vee q) \wedge r$

p	q	r	$p \vee q$	$\neg (p \vee q)$	$\neg (p \vee q) \wedge r$
T	T	T	T	F	F
T	T	F	T	F	F
T	F	T	T	F	F
T	F	F	T	F	F
F	T	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T
F	F	F	F	T	F

Inverse, Converse, and Contrapositive

Proposition: $p \rightarrow q$

If you build it, they will come

Definition: Inverse: $\neg p \rightarrow \neg q$

If you don't build it, they will not come

Definition: Converse: $q \rightarrow p$

If they come, then you built it

Definition: Contrapositive: $\neg q \rightarrow \neg p$

If they don't come, you didn't build it

Inverse, Converse, and Contrapositive

				(Orig.)	(Contrapos.)	(Inverse)	(Converse)
r	s	$\neg r$	$\neg s$	$r \rightarrow s$	$\neg s \rightarrow \neg r$	$\neg r \rightarrow \neg s$	$s \rightarrow r$
T	T	F	F				
T	F	F	T				
F	T	T	F				
F	F	T	T				

Thus, Original $\equiv$ Contrapositive

Inverse $\equiv$ Converse

Inverse, Converse, and Contrapositive

				(Orig.)	(Contrapos.)	(Inverse)	(Converse)
r	s	$\neg r$	$\neg s$	$r \rightarrow s$	$\neg s \rightarrow \neg r$	$\neg r \rightarrow \neg s$	$s \rightarrow r$
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	T
F	T	T	F	T	T	F	F
F	F	T	T	T	T	T	T

(Original) If UA is in session, beer sales increase.

(Contrapos.) If beer sales don't increase, UA isn't in session.

(Converse) If beer sales increase, UA is in session.

(Inverse) If UA isn't in session, beer sales do not increase.

Contraposition

An implication $p \rightarrow q$ gives you interesting information in two scenarios:

- If p is TRUE, then q must be TRUE
- If q is FALSE, then p must be FALSE

.

More on Propositions

- Are the following propositions (statements)?
- Perry is a Platypus.
- Yes (it happens to be true)
- Perry is a Duck.
- Yes (it happens to be false)

More on Propositions

- Are the following propositions (statements)?
- Where's Perry?
- No
- Why not?
- It's a question.

More on Propositions

- Are the following propositions (statements)?
- Get on the rollercoaster.
- No
- Why not?
- It's a command.

More on Propositions

- Are the following propositions (statements)?
- $x + 5 = 7$
- No
- Why not?
- It contains a variable and so it can be true or false.

More on Propositions

- Are the following propositions (statements)?
- This is a statement.
- Yes (it happens to be true)
- This is not a statement.
- Hmm, it seems simple but ...

More on Propositions

- “This is not a statement.”
- Suppose it’s false.
- Then it is a statement, which means it must be true.
- But we just said it’s false! That can’t happen!
- OK, then it’s true.
- That means it is not a statement, but a statement is something that is true or false, so it IS a statement.

More on Propositions

- “This is not a statement.”
- This is an example of what is called a paradox, something that looks like a statement but can’t be true or false.
- Paradoxes look weird, but we don’t need to stress about them.

Paradoxes

- You may have seen the old Star Trek episode where they defeat a computer with something like:
- Kirk: Everything Spock says is a lie.
- Spock: I am lying.
- Computer: If you are lying then you must be telling the truth, but then you are telling a lie, which means you are telling the truth, which means you are telling a lie . . .
- and smoke comes out of the computer that apparently wasn't programmed to recognize paradoxes.

Paradoxes

- Another famous paradox is:
One barber lives in a small town. In that town the barber cuts the hair of everyone who doesn't cut their own hair and doesn't cut the hair of those who do cut their own hair.
- Who cuts the barbers hair?
- How can we possibly resolve this?
- Don't panic. It just means no such barber can exist.

Can a False Hypothosis Prove Anything?

If I am a little green man from Mars, then the Earth will explode at the end of this class.

Can a False Hypothesis Prove Anything?

If I am a little green man from Mars, then the Earth will explode at the end of this class.

g: I am a little green man from Mars

e: The Earth will explode at the end of this class

$$g \rightarrow e$$

Can a False Hypothesis Prove Anything?

If I am a little green man from Mars, then the Earth will explode at the end of this class.

g: I am a little green man from Mars

e: The Earth will explode at the end of this class

$$g \rightarrow e$$

g is F

$g \rightarrow e$ is T

e is undetermined

The statement is nonsense

Recap – Logical Operators

p	$\neg p$
T	F
F	T

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Bit Operations

- Computers store information in bits
- A bit is off or on, which we represent as 0 or 1
- Since this has two values, we can also think of these values as false (0) or true(1)
- Often, we represent Boolean expressions using 0 and 1 instead of true/false

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Bit Operations

- A string of bits, is a bit string. Most computer languages let you use Boolean operators on bit strings.
- For example, if $x = 01011100$ and $y = 11010110$
- Then in C,
 $z = x \mid y;$
stores the bitwise or of these two strings (bytes)
- e.g., after the command $z = 11011110$

1.2 Applications of Propositional Logic

Translating English Statements

You can take 144 only if you have completed 110 unless you have advisor approval.

First, assign variables:

t: take 144

c: completed 110

a: instructor approval

Does this work?

$c \vee a \rightarrow t$

NO! This is says you can take the class if you've met the conditions, but not that the conditions are required.

1.2 Applications of Propositional Logic

Translating English Statements

You can take 144 only if you have completed 110 unless you have advisor approval.

First, assign variables:

t: take 144

c: completed 110

a: advisor approval

$$t \rightarrow (c \vee a)$$

This is correct. If you're in the class, then it must be true that you've completed 110 or got approval.

1.2 Applications of Propositional Logic

Translating English Statements

You cannot ride the rollercoaster if you are under 4 feet tall unless you are older than 16 years.

First, assign variables:

r: ride the rollercoaster

u: under 4 feet tall

o: older than 16

$$(u \wedge \neg o) \rightarrow \neg r$$

1.2 Applications of Propositional Logic

System Specifications

Requirements for systems are often expressed in English. They must be interpreted as logic. Then we must make sure we can satisfy the conditions.

1.2 Applications of Propositional Logic

System Specifications

1. The diagnostic message is stored in the buffer or it is retransmitted.
2. The diagnostic message is not stored in the buffer.
3. If the diagnostic message is stored in the buffer, then it is retransmitted.

1. The diagnostic message is stored in the buffer or it is retransmitted.
2. The diagnostic message is not stored in the buffer.
3. If the diagnostic message is stored in the buffer, then it is retransmitted.

b: diagnostic message stored in buffer

r: message is retransmitted

$b \vee r$

$\neg b$

$b \rightarrow r$

We want to know if we can satisfy the specification. We can if the specification is consistent. This means there's an assignment that make all specifications true.

1. The diagnostic message is stored in the buffer or it is retransmitted.
2. The diagnostic message is not stored in the buffer.
3. If the diagnostic message is stored in the buffer, then it is retransmitted.

b : diagnostic message stored in buffer

r : message is retransmitted

$b \vee r$

$\neg b$

$b \rightarrow r$

b	r	$b \vee r$	$\neg b$	$b \rightarrow r$
T	T	T	F	T
T	F	T	F	F
F	T	T	T	T
F	F	F	T	T

1. The diagnostic message is stored in the buffer or it is retransmitted.
2. The diagnostic message is not stored in the buffer.
3. If the diagnostic message is stored in the buffer, then it is retransmitted.

b : diagnostic message stored in buffer

r : message is retransmitted

$b \vee r$

$\neg b$

$b \rightarrow r$

$b = \text{False}$

$r = \text{True}$

Works



b	r	$b \vee r$	$\neg b$	$b \rightarrow r$
T	T	T	F	T
T	F	T	F	F
F	T	T	T	T
F	F	F	T	T

1. The diagnostic message is stored in the buffer or it is retransmitted.
2. The diagnostic message is not stored in the buffer.
3. If the diagnostic message is stored in the buffer, then it is retransmitted.

What if we add the condition

4. The diagnostic message is stored in the buffer.

Now we are no longer consistent
because our only possible solution no
longer works.

b = False
r = True

1.2 Applications of Propositional Logic

Logic Puzzles

There are a series of puzzles based on a mythical island of knights and knaves. Knights always tell the truth. Knaves always lie. The goal is to figure out who is a knight and who is a knave based on what they say.

For Example:

You meet two characters, Doof and Perry.

Doof says: “At least one of is a knave.”

Perry says nothing.

Can you tell what each character is?

How do you know?

Knights and Knaves

Doof says: “At least one of is a knave.”

Perry says nothing.

D: D is knight

P: P is knight

$$D \rightarrow \neg D \vee \neg P$$

$$\neg D \rightarrow D \wedge P$$

D	P	$\neg D$	$\neg P$	$\neg D \vee \neg P$	$D \wedge P$	$D \rightarrow \neg D \vee \neg P$	$\neg D \rightarrow D \wedge P$
T	T	F	F	F	T	F	T
T	F	F	T	T	F	T	T
F	T	T	F	T	F	T	F
F	F	T	T	T	F	T	F

So Doof is a knight and Perry is a knave is consistent.

Doof says: “At least one of is a knave.”

Perry says nothing.

Another way to think of it:

If Doof is knight then what he says is true so Perry must be a knave. Perry says nothing so there is no contradiction there.

But we have to make sure this is the only possible solution.

If Doof is a knave then what he says is false, so there are no knaves, so Doof is a knight $\Rightarrow \Leftarrow$ (contradiction)

Therefore, Doof is a knight and Perry is a knave.

1.2 Applications of Propositional Logic

Logic Circuits



Inverter

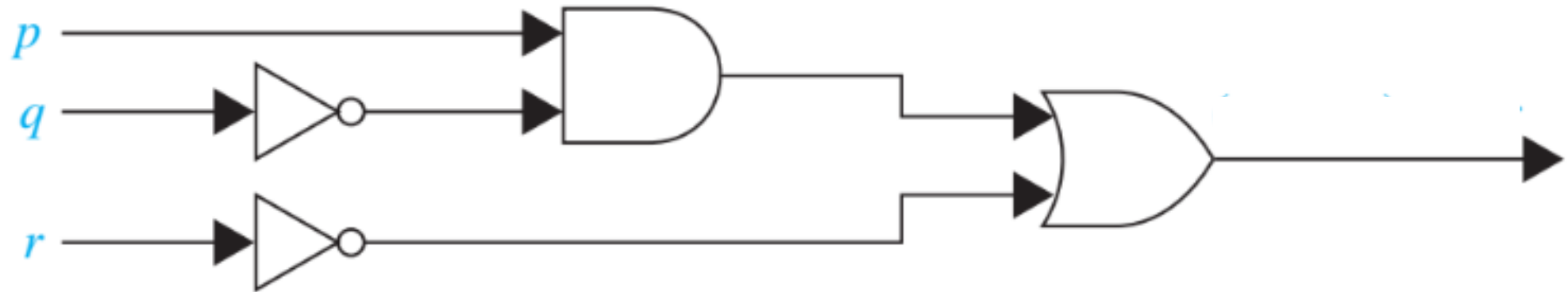


OR gate



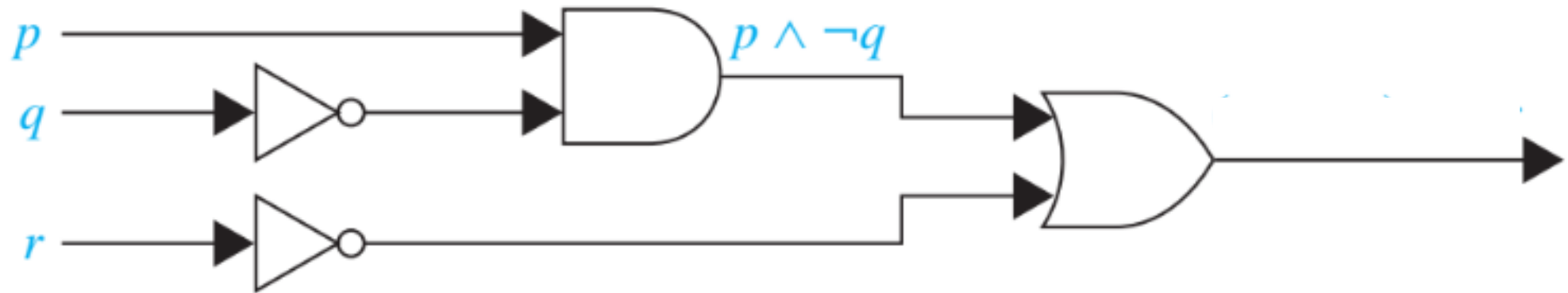
AND gate

Logic Circuits



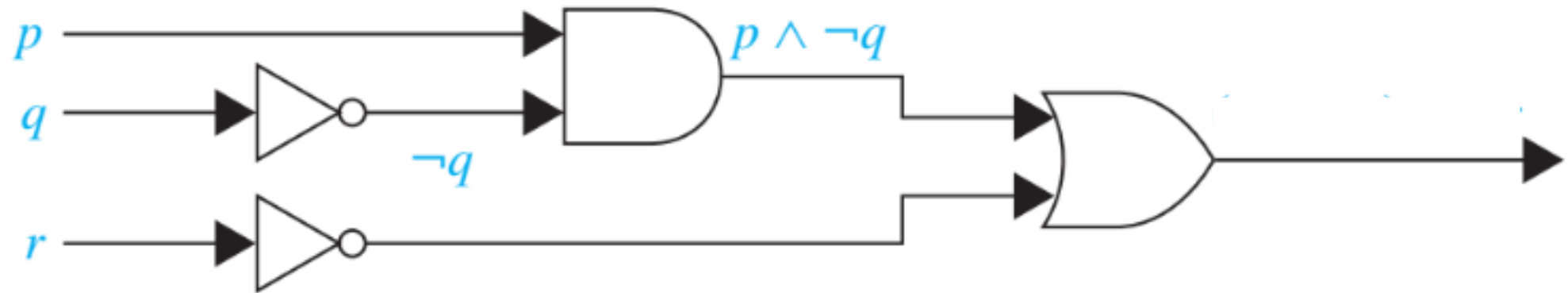
Determine the output for the combinatorial circuit.

Logic Circuits



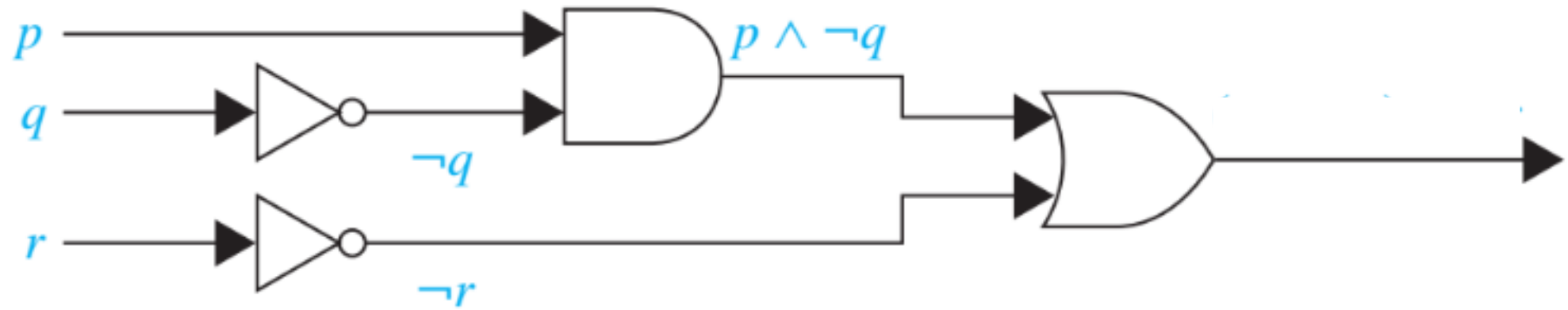
Determine the output for the combinatorial circuit.

Logic Circuits



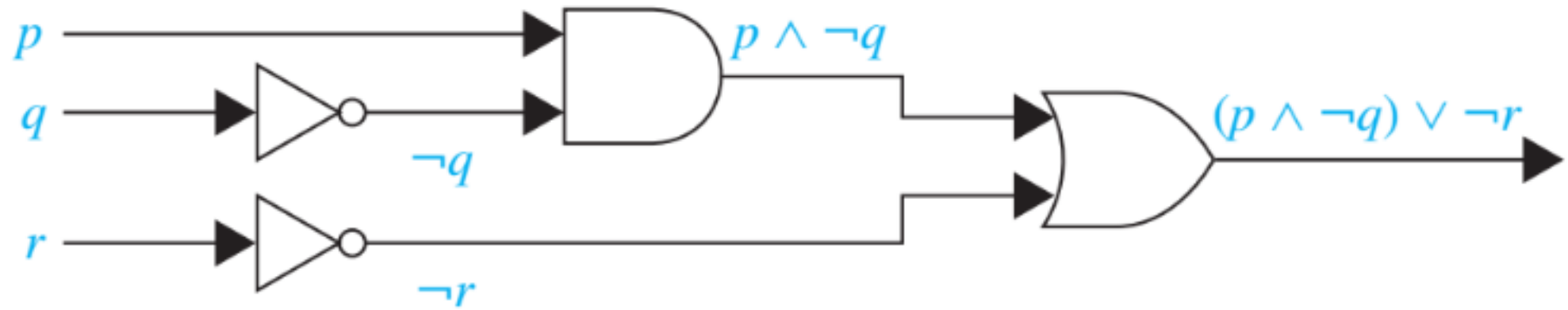
Determine the output for the combinatorial circuit.

Logic Circuits



Determine the output for the combinatorial circuit.

Logic Circuits



Determine the output for the combinatorial circuit.

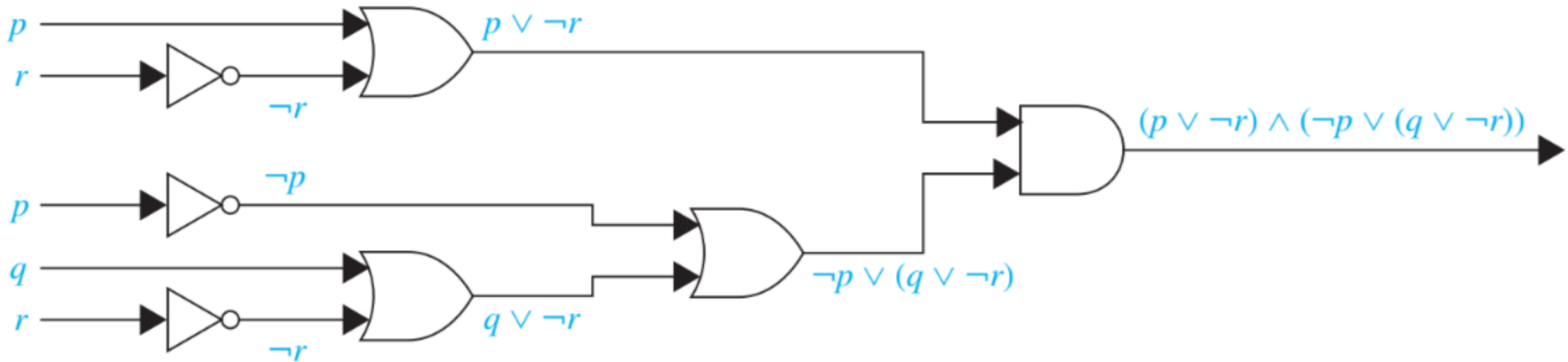


Logic Circuits

Build a digital circuit that produces the output $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$.

Logic Circuits

Build a digital circuit that produces the output $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$.



End of Section 1.2
Starting Section 1.3

Two Categories of Propositions

Definition: Tautology

A proposition that always evaluates to true.

Definition: Contradiction

A proposition that always evaluates to false.

Two Categories of Propositions

Definition: Tautology

A proposition that always evaluates to true.

examples: T , $p \vee \neg p$

Definition: Contradiction

A proposition that always evaluates to false.

examples: F , $p \wedge \neg p$

Determining Tautologies

How can we determine whether a proposition is a tautology or a contradiction?

One way is to use a truth table.

To show something IS a tautology this way one would have to have complete truth table that shows the proposition evaluates to true for all possible combinations of values of the logical variables.

To show it's NOT a tautology one only has to find one assignment of variables under which the proposition evaluates to false

(Likewise for a contradiction)

For example, show whether

$$p \oplus (q \wedge r)$$

is a tautology or a contradiction or neither?

$$F \oplus (F \wedge F) = F \oplus F = F$$

so not a tautology

$$T \oplus (F \wedge F) = T \oplus F = T$$

so not a contradiction

Satisfiable Proposition

A proposition is ***Satisfiable*** if there is an assignment of truth values to its variables that makes it true.

In other words: a proposition is satisfiable iff it is not a contradiction.

Example: Show $(p \wedge q) \rightarrow \neg p$ is satisfiable

we must find an assignment of values for p and q that make the statement true

This is easy since all we need to do is make the hypothesis false

p = true and q = false will work.

$$t \wedge f \rightarrow \neg t \equiv f \rightarrow f \equiv t$$

Biconditionals and Logical Equivalence

Definition: Logically Equivalent

Compound propositions p and q are logically equivalent ($p \equiv q$) if $p \leftrightarrow q$ is a tautology.

In other words, the truth tables for p and q are the same.

Showing Propositions are Logically Equivalent

Just like we can use truth tables to show a proposition is a tautology, we can also use truth tables to show two propositions are equivalent.

For example, are the propositions

$p \rightarrow q$ and
 $\neg p \vee q$ equivalent?

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T			
T	F			
F	T			
F	F			

Showing Propositions are Logically Equivalent

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Showing Propositions are Logically Equivalent

Just like we can use truth tables to show a proposition is a tautology, we can also use truth tables to show two propositions are equivalent.

For example, are the propositions

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and

$\neg p \vee q$

equivalent?

Yes!

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Is Exclusive Or Associative?

p	q	r	$p \oplus q$	$q \oplus r$	$(p \oplus q) \oplus r$	$p \oplus (q \oplus r)$
T	T	T	F	F	T	T
T	T	F	F	T	F	F
T	F	T	T	T	F	F
T	F	F	T	F	T	T
F	T	T	T	F	F	F
F	T	F	T	T	T	T
F	F	T	F	T	T	T
F	F	F	F	F	F	F

So: $(p \oplus q) \oplus r \equiv p \oplus (q \oplus r)$

And we can write $p \oplus q \oplus r$ without ambiguity

So Are All Our Operators Associative?

Let's do a quiz to figure out if and, or, and implication are also associative.

De Morgan's Laws

$$(1) \overline{p \wedge q} \equiv \overline{p} \vee \overline{q}$$

$$(2) \overline{p \vee q} \equiv \overline{p} \wedge \overline{q}$$

Example(s):

$$\begin{aligned} \overline{a \vee (b \vee c)} &\equiv \overline{a} \wedge \overline{(b \vee c)} \\ &\equiv \overline{a} \wedge (\overline{b} \wedge \overline{c}) \end{aligned}$$

De Morgan's Laws

So we see that DeMorgan's laws generalize.

$$\overline{p_1 \vee p_2 \vee \cdots \vee p_n} \equiv \overline{p_1} \wedge \overline{p_2} \wedge \cdots \wedge \overline{p_n}$$

$$\overline{p_1 \wedge p_2 \wedge \cdots \wedge p_n} \equiv \overline{p_1} \vee \overline{p_2} \vee \cdots \vee \overline{p_n}$$

More DeMorgan

The defendants who've paid their fine or scheduled an appeal are OK. We want to report the others.

More DeMorgan

The defendants who've paid their fine or scheduled an appeal are OK. We want to report the others.

So say

p: is paid fine

s: is scheduled appeal

Your fellow programmer has written

if (not p or not s):

 #report them

Do you need to fix the code?

More DeMorgan

The defendants who've paid their fine or scheduled an appeal are OK. We want to report the others.

So say

p: is paid fine

s: is scheduled appeal

Your fellow programmer has written

if (not p or not s):

 #report them

Do you need to fix the code? **Yes! $\text{not}(p \text{ or } s) = \text{not } p \text{ and not } s$**

More DeMorgan

**Suppose we were skeptical that DeMorgan's laws work.
How can we demonstrate that they are correct?**

Construct truth tables!

Laws of Propositional Logic

We can use known logical equivalences to simplify propositions.
Here are a few of the named equivalences:

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Laws of Propositional Logic

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (1)

By use of a Truth Table;

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Because the expression evaluates to true for all possible arrangements of truth values, the expression is a tautology.

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (2)

By application of logical equivalences:

$$(p \wedge q) \rightarrow p \equiv \neg (p \wedge q) \vee p$$

$$\equiv \neg p \vee \neg q \vee p$$

$$\equiv \neg p \vee p \vee \neg q$$

$$\equiv p \vee \neg p \vee \neg q$$

$$\equiv T \vee \neg q$$

$$\equiv \neg q \vee T$$

$$\equiv T$$

- Conditional Identities

-DeMorgan's Laws

-Commutative Laws

-Commutative Laws

-Complement Laws

-Commutative Laws

-Domination Laws

This is correct, but I personally think this is more steps than we need.

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (2)

By application of logical equivalences:

$$(p \wedge q) \rightarrow p \equiv \neg (p \wedge q) \vee p$$

$$\equiv \neg p \vee \neg q \vee p$$

$$\equiv \neg p \vee p \vee \neg q$$

$$\equiv T \vee \neg q$$

$$\equiv T$$

- Conditional Identities

-DeMorgan's Laws

-Commutative Laws

-Complement Laws

-Domination Laws

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (3)

By Reasoning:

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (3)

By Reasoning:

When p is true:

$(T \wedge q) \rightarrow T$ is true regardless of q

(Why? By $\rightarrow$'s truth table, $? \rightarrow T \equiv T$!)

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (3)

By Reasoning:

When p is true:

$(T \wedge q) \rightarrow T$ is true regardless of q

(Why? By $\rightarrow$'s truth table, $? \rightarrow T \equiv T$!)

When p is false:

$(F \wedge q) \rightarrow F$ is also true regardless of q

(Why? $F \wedge q \equiv F$, and $F \rightarrow F \equiv T$!)

Applications of Logical Equivalences

Question: Is $(p \wedge q) \rightarrow p$ a tautology? (3)

By Reasoning:

When p is true:

$(T \wedge q) \rightarrow T$ is true regardless of q

(Why? By $\rightarrow$'s truth table, $? \rightarrow T \equiv T$!)

When p is false:

$(F \wedge q) \rightarrow F$ is also true regardless of q

(Why? $F \wedge q \equiv F$, and $F \rightarrow F \equiv T$!)

Thus, $(p \wedge q) \rightarrow p$ is a tautology

What we just learned

Three quick ways to prove that something is a **tautology**:

What we just learned

Three quick ways to prove that something is a **tautology**:

- Truth Table
 - Do all cases resolve to TRUE ?

What we just learned

Three quick ways to prove that something is a **tautology**:

- Truth Table
 - Do all cases resolve to TRUE ?
- Logical Equivalences
 - Can we convert the expression to TRUE ?

What we just learned

Three quick ways to prove that something is a **tautology**:

- Truth Table
 - Do all cases resolve to TRUE ?
 - Logical Equivalences
 - Can we convert the expression to TRUE ?
 - Reasoning
 - Any argument you make
 - Our example did “proof by cases”
-

Proving that something is a contradiction?

How to prove that something is a **contradiction**?

- Truth Table
 - Do all cases resolve to FALSE ?
 - Logical Equivalences
 - Can we convert the expression to FALSE ?
 - Reasoning
 - Any argument you make
-

Applications of Logical Equivalences

Example(s): [Assume `games` is an integer value.]

```
if ( (games <= 10 or ties > 2) and games >= 11 ) :  
Let g:games <= 10           t:ties > 2           ¬g: games >= 11
```

Applications of Logical Equivalences

Example(s): [Assume `games` is an integer value.]

```
if ((games <= 10 or ties > 2) and games >= 11) :  
Let g:games <= 10           t:ties > 2           ¬g: games >= 11
```

$(g \vee t) \wedge \neg g \equiv (g \wedge \neg g) \vee (t \wedge \neg g)$ [Distribution]

$\equiv \mathbf{F} \vee (t \wedge \neg g)$ [Negation]

$\equiv t \wedge \neg g$ [Identity]

Applications of Logical Equivalences

Example(s): [Assume `games` is an integer value.]

```
if ((games <= 10 or ties > 2) and games >= 11) :  
Let g:games <= 10          t:ties > 2          ¬g: games >= 11
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$(g \vee t) \wedge \neg g \equiv (g \wedge \neg g) \vee (t \wedge \neg g)$ [Distribution]

$\equiv \mathbf{F} \vee (t \wedge \neg g)$ [Negation]

$\equiv t \wedge \neg g$ [Identity]

Thus, the statement can be rewritten more efficiently as:

```
if (ties > 2 and games >= 11) :
```

Satisfiability

- The book has a couple of examples of applications (Placing Queens, Sudoku) you may want to look at.
- How hard is it to tell if logical expression is satisfiable?
- If we have n variables, how many rows does the truth table have?
- 2^n
- This is called exponential growth . . . and it is bad.

Satisfiability

- If we had 100 variables, and could fill in 1,000 rows a second, it would take ~ 42 quintillion years to finish. That's 3 billion times longer than the universe has existed. (they think)
- Is there a smarter (faster) way to determine if a general Boolean expression is satisfiable?
- We don't think so.
- The satisfiability problem has historic importance since it was the first problem proved to be "NP-Complete". Which roughly means it is at least as hard as any of a wide class of problems.

End of Section 1.3
Starting Section 1.4

Predicates and Quantifiers

Definition: Predicate (a.k.a. Propositional Function)

A statement that includes at least one variable and will evaluate to either true or false when the variable(s) are assigned values.

Examples:

$P(x): x^2 = 4$ $P(-2)$ is true $P(3)$ is false

$Q(x)$: x is a student in 144

$R(x, y, z)$: x is the parent of y and z

Predicates and Quantifiers

The domain of a variable in a predicate is the set of all possible values for the variable.

Examples:

$P(x): x^2 = 4$ domain might be all integers

or all positive integers

or all real numbers

or even $\{0, 5, 7\}$

How do we know what the domain is?

We must be told.

Quantification

Idea: Establish truth of predicates over sets of values.

Two common generalizations:

(1) Universal Quantification ($\forall$)

– Reads “for every” or “for all”

(2) Existential Quantification ($\exists$)

– Reads “There exists”

$\exists!$ is another quantification

– Reads “There exists a unique”

$\forall x \ P(x)$ is true only when $P(x)$ is true for each x in the domain. (False otherwise.)

(easy to prove false, hard to prove true)

$\exists x \ P(x)$ is true when at least one domain member causes $P(x)$ to evaluate to true.

(easy to prove true, hard to prove false)

Another view

Suppose the domain of a predicate $P(x)$ is $\{a_1, a_2, \dots, a_n\}$

$$\forall x P(x) \equiv P(a_1) \wedge P(a_2) \wedge \dots \wedge P(a_n)$$

$$\exists x P(x) \equiv P(a_1) \vee P(a_2) \vee \dots \vee P(a_n)$$

Examples

Suppose $P(x)$ is: x is a student named Dylan

Is $P(x)$ a proposition?

No, because whether it is true or false depends on a variable.

So is $\forall x P(x)$ a proposition?

Yes, but . . . the domain of P must be defined

Say the domain of P is all students in this class

Is $\forall x P(x)$ true or false?

Is $\exists x P(x)$ true or false?

Is $\exists! x P(x)$ true or false?

Examples

P(x): x has feathers and the domain is birds

Let's translate all the quantifiers to English

$$\forall x P(x)$$

Every bird has feathers

$$\exists x P(x)$$

There is a bird that has feathers

Quantified Statements

We can add quantifiers to compound propositions:

$S(x)$: x is a secret agent

$P(x)$: x is a platypus

Domain: all real life and fictional characters

$\exists x (S(x) \wedge P(x))$

There is a secret agent who is also a platypus

This is true.

$\exists x (S(x) \wedge \neg P(x))$

There is a secret agent who is not a platypus

This is also true.

Quantified Statements

A proposition that includes a quantifier is called a quantified statement

Note that logical operators have higher priority than quantifiers.

so $\exists x S(x) \wedge P(x)$

is evaluated as $(\exists x S(x)) \wedge P(x)$

This not a proposition.

The variable in $S(x)$ is called a bound variable
because it's bound to a quantifier.

The variable in $P(x)$ is called a free variable

Usually, we would use different labels for these variables to avoid confusion.

e.g. $\exists x S(x) \wedge P(y)$

Quantified Statements

The part of a logical expression to which a quantifier is applied is called the scope of that quantifier.

For example in:

$$\forall x (P(x) \rightarrow Q(y)) \wedge R(x)$$

the scope of $\forall x$ is $P(x) \rightarrow Q(y)$

Examples

$C(x)$ x has a cape

$F(x)$ x can fly

$B(x)$ x is bullet proof

The domain is superheroes

Express

There is a bullet proof superhero who can fly and has a cape.

Every flying superhero has a cape.

Every superhero flies, is bullet proof, or has a cape.

A superhero can fly or be bullet proof, but not both.

$C(x)$ x has a cape

$F(x)$ x can fly

$B(x)$ x is bullet proof

The domain is superheroes

Express

There is a bullet proof superhero who can fly and has a cape.

$$\exists x (C(x) \wedge F(x) \wedge B(x))$$

Every flying superhero has a cape.

Every superhero flies, is bullet proof, or has a cape.

A superhero can fly or be bullet proof, but not both.

$C(x)$ x has a cape

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$$\exists x (C(x) \wedge F(x) \wedge B(x))$$

Every flying superhero has a cape.

$$\forall x (F(x) \rightarrow C(x))$$

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A superhero can fly or be bullet proof, but not both.

$C(x)$ x has a cape

$F(x)$ x can fly

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$$\exists x (C(x) \wedge F(x) \wedge B(x))$$

Every flying superhero has a cape.

$$\forall x (F(x) \rightarrow C(x))$$

Every superhero flies, is bullet proof, or has a cape.

$$\forall x (F(x) \vee B(x) \vee C(x))$$

A superhero can fly or be bullet proof, but not both.

$C(x)$ x has a cape

$F(x)$ x can fly

$B(x)$ x is bullet proof

The domain is superheroes

Express

There is a bullet proof superhero who can fly and has a cape.

$$\exists x (C(x) \wedge F(x) \wedge B(x))$$

Every flying superhero has a cape.

$$\forall x (F(x) \rightarrow C(x))$$

Every superhero flies, is bullet proof, or has a cape.

$$\forall x (F(x) \vee B(x) \vee C(x))$$

A superhero can fly or be bullet proof, but not both.

$$\forall x (\neg(F(x) \wedge B(x)))$$

So $P(x)$: x is red, domain is types of Birds
is a function, it is not true or false as it is

$P(\text{blue jay})$ blue jay is red Does evaluate to T or F

Adding quantifiers also makes the statement evaluate to T or F

$\exists x P(x)$ "There is a red bird" also evaluates to T or F

Examples

$P(x)$ is $x + 1 = 2$

is

$\exists x P(x)$ true ?

Examples

$P(x)$ is $x + 1 = 2$

is

$\exists x P(x)$ true ?

We don't know! What is the domain?

If the domain is all integers, then it is true.

If the domain is all even integers, then it is false.

Examples

$P(x)$ is $x + 1 = 2$

Domain is all integers.

is

$\exists x P(x)$ true ?

Yes! ($x = 1$)

Examples

$P(x)$ is $x + 1 = 2$

Domain is all integers.

is

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Yes! ($x = 1$)

how about

$\forall x P(x)$?

Examples

$P(x)$ is $x + 1 = 2$

Domain is all integers.

is

$\exists x P(x)$ true ?

Yes! ($x = 1$)

how about

$\forall x P(x)$?

False. ($x = 0$)

Example: Universal Quantification

Consider this conversational English statement:

All of Dr. Anson's students are geniuses.

How can we express that statement in logic notation?

Example: Universal Quantification

Consider this conversational English statement:

All of Dr. Anson's students are geniuses.

How can we express that statement in logic notation?

Easiest way: Let Domain be Dr. Anson's students

Define $P(x)$ = x is a genius

$\forall x P(x)$

Example: Universal Quantification

Consider this conversational English statement:

All of Dr. Anson's students are geniuses.

How can we express that statement in logic notation?

If we must use the domain All People

Define $P(x)$ = x is a genius

Define $Q(x)$ = x is a student of Dr. Anson

$\forall x (Q(x) \rightarrow P(x))$

More DeMorgan

Suppose $P(x)$ is a predicate with domain $\{a_1, a_2, \dots, a_n\}$

$$\neg \forall x P(x) \equiv \neg(P(a_1) \wedge P(a_2) \wedge \dots \wedge P(a_n))$$

$$\equiv \neg P(a_1) \vee \neg P(a_2) \vee \dots \vee \neg P(a_n)$$

$$\equiv \exists x \neg P(x)$$

DeMorgan also works if the domain is infinite, so:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

More DeMorgan

Suppose $P(x)$ is a predicate with domain $\{a_1, a_2, \dots, a_n\}$

$$\neg \exists x P(x) \equiv \neg(P(a_1) \vee P(a_2) \vee \dots \vee P(a_n))$$

$$\equiv \neg P(a_1) \wedge \neg P(a_2) \wedge \dots \wedge \neg P(a_n)$$

$$\equiv \forall x \neg P(x)$$

DeMorgan also works if the domain is infinite, so:

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

More DeMorgan – in English

Not every secret agent is human

$\neg \forall x P(x)$ $P(x)$: x is human, domain all secret agents

There is a secret agent who is not a human.

$\exists x \neg P(x)$

There is no rollercoaster that is not fun.

$\neg \exists x \neg P(x)$ $P(x)$: x is fun, domain all rollercoasters

Every rollercoaster is fun.

$\forall x P(x)$ $\equiv \forall x \neg \neg P(x)$

Predicates with Multiple Variables

So far, all the predicates we've looked at have a single variable, but there's no reason we can't have multiple variables.

Examples:

$P(x,y)$: x and y are brothers

We need domains for both vars. Here the domains for x and y might be the same.

$Q(x,y)$: x is covered with y

The domain for x is all animals.

The domain for y is {scales, fur, feathers}

$Q(\text{fox}, \text{fur})$ is true, $Q(\text{lizard}, \text{feathers})$ is false

Of course, we could also have 3, 4, or even 255 variables. (i.e. any number)

End of Section 1.4
Starting Section 1.5

Quantifiers with Multiple Variables

All the variables must be bound to make a predicate a proposition.

Examples:

$\forall x \exists y P(x,y)$ is a proposition (both x and y are bound)

$\exists x \exists y \exists z Q(x, y, z)$ is also a propositions (all vars bound)

$\forall x P(x,y)$ is a not proposition (y is free, so truth depends on value of y)

Combining Quantifiers

- $\exists x \exists y P(x,y)$ There exists an x and a y such that $P(x,y) = T$
- $\exists x \forall y P(x,y)$ There exists an x s.t. for every y $P(x,y) = T$
- $\forall x \exists y P(x,y)$ For every x there exists a y s.t. $P(x,y) = T$
- $\forall x \forall y P(x,y)$ For every x and for every y $P(x,y) = T$

Combining Quantifiers

- $\exists x \exists y P(x,y)$ There exists an x and a y such that $P(x,y) = T$
- $\exists x \forall y P(x,y)$ There exists an x s.t. for every y $P(x,y) = T$
- $\forall x \exists y P(x,y)$ For every x there exists a y s.t. $P(x,y) = T$
- $\forall x \forall y P(x,y)$ For every x and for every y $P(x,y) = T$

Suppose $P(x,y)$ is x eats y , $x \in \text{Mammals}$, $y \in \text{Insects}$

Then $\exists x \exists y P(x,y)$ in plain English reads “There is a mammal that eats some insect.”

$\exists x \forall y P(x,y)$ reads “Some mammal eats every insect.”

$\forall x \exists y P(x,y)$ reads “Every mammal eats some insect”

$\forall x \forall y P(x,y)$ reads “Every mammal eats every insect”

Can have even more: $\forall x \exists y \forall z (P(x,y) \rightarrow P(x,z))$

More Nested Quantifier Practice

Let $P(x, y)$ x teaches y

where the domain is UA Faculty $\times$ UA Courses

Write the expression for Dr. Anson teaches exactly 2 courses

More Nested Quantifier Practice

Let $P(x, y)$ x teaches y

where the domain is UA Faculty $\times$ UA Courses

Write the expression for Dr. Anson teaches exactly 2 courses

$$\exists w \exists z \forall y (P(\text{Anson}, w) \wedge P(\text{Anson}, z) \wedge (w \neq z) \wedge \\ (P(\text{Anson}, y) \rightarrow ((y = w) \vee (y = z))))$$

$$w, z, y \in \text{UA Courses}$$

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

Easiest

Domain is students in this class.

$P(x)$ is owns a cat and a dog.

- $\exists x P(x)$

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

Easiest

Domain is students in this class.

$P(x)$ is owns a cat and a dog.

- $\exists x P(x)$

Now let's require we use a different domain.

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

New domain, must contain all students in this class.

Domain is students at U of A.

$P(x)$ is owns a cat and a dog.

$C(x)$ x is in this class

$\exists x (C(x) \wedge P(x))$

Now add the requirement of using a predicate with two variables.

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

$P(x, y)$ is x owns y , $x \in \text{Students at U of A}$, $y \in \text{Animals}$

$C(x)$ x is in this class, $x \in \text{Students at U of A}$

$\exists x (C(x) \wedge P(x, \text{cat}) \wedge P(x, \text{dog}))$

More Examples

Everybody loves somebody.

$$\forall x \exists y P(x, y)$$

$P(x, y)$: x loves y

There is somebody who is loved by everyone.

$$\exists y \forall x P(x, y)$$

Order of quantifiers is important!

DeMorgan for Multiple Quantifiers

Recall:

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

This idea carries through for multiple quantifiers.

$$\neg \exists x \forall y P(x, y) \equiv \forall x \neg \forall y P(x, y) \equiv \forall x \exists y \neg P(x, y)$$

DeMorgan for Multiple Quantifiers

example: $P(x, y)$: x loves y Domain all people

$\forall x \exists y P(x, y)$ Everyone loves someone

$\neg \forall x \exists y P(x, y)$

$\exists x \forall y \neg P(x, y)$ There is someone who doesn't love anyone

Everyone Else

Suppose we have a domain of students in an art class. We want to express the sentence "Everyone in the class painted a portrait of someone else in the class."

We might try something like:

$M(x,y)$: x painted a portrait of y

$\forall x \exists y M(x, y)$

The problem is, this might evaluate to true, even if some student, Narcissus, only painted a self portrait. (i.e. $M(\text{Narcissus}, \text{Narcissus})$ is true, but $M(\text{Narcissus}, y)$ is false for all other students y .)

How do we capture the "someone else" part of the proposition?

Everyone Else

Suppose we have a domain of students in an art class. We want to express the sentence "Everyone in the class painted a portrait of someone else in the class."

How about:

$M(x,y)$: x painted a portrait of y

$\forall x \exists y ((x \neq y) \wedge M(x,y))$

This works.

Everyone Else

Suppose we have a domain of students in an art class. Now suppose we want to express the sentence "Everyone in the class painted a portrait of everyone else in the class." Notice that this can be true even if no one paints a self portrait.

As before, we can't do:

$M(x,y)$: x painted a portrait of y

$\forall x \forall y M(x,y)$

Since everyone would have to paint a portrait of themselves for this to be true.

Everyone Else

Suppose we have a domain of students in an art class. Now suppose we want to express the sentence "Everyone in the class painted a portrait of everyone else in the class." Notice that this can be true even if no one paints a self portrait.

Last time we fixed our problem by adding the condition x is not equal to y :

$M(x,y)$: x painted a portrait of y

$\forall x \forall y ((x \neq y) \wedge M(x, y))$

Does this fix our problem in this case?

No, this makes the statement false for almost all domains since every student equals themselves.

What is the one special case when the statement would be true?

Everyone Else

Suppose we have a domain of students in an art class. Now suppose we want to express the sentence "Everyone in the class painted a portrait of everyone else in the class." Notice that this can be true even if no one paints a self portrait.

We need an implication for this:

$M(x,y)$: x painted a portrait of y

$\forall x \forall y ((x \neq y) \rightarrow M(x, y))$

Now it has to be true that every paints a portrait of everyone else?

There exists a unique

Recall $\exists!x P(x)$ means there's exactly one element in the domain that makes $P(x)$ true.

How can we express this idea in logic?

$$\exists x \forall y (P(x) \wedge (P(y) \rightarrow (x = y)))$$

or

$$\exists x \forall y (P(x) \wedge (x \neq y) \rightarrow \neg P(y))$$

Moving Quantifiers

You may move a quantifier through a logical statement as long as you don't cross another quantifier or a variable of the same name.

For example, suppose we have the predicates:

$P(x)$: x is a secret agent

$D(y, x)$: y is the nemesis of x (The domain of both is all characters)

We can express the statement "every secret agent has a nemesis" as:

$$\forall x (P(x) \rightarrow \exists y D(y, x))$$

which is equivalent to

$$\forall x \exists y (P(x) \rightarrow D(y, x))$$

Moving Quantifiers

You may move a quantifier through a logical statement as long as you don't cross another quantifier or a variable of the same name. Also, if you cross a negation, you must use DeMorgan.

For example, suppose we have the predicates:

$P(x)$: x is a secret agent

$D(y, x)$: y is the nemesis of x

We can express the statement "every secret agent has a nemesis" as:

$$\forall x \exists y (P(x) \rightarrow D(y, x))$$

however this is not equivalent to

$$\exists y \forall x (P(x) \rightarrow D(y, x))$$

which means "there is someone who is the nemesis of all secret agents"

Moving Quantifiers

You actually can cross another quantifier, as long as it is of the same type.

$P(x)$: x is a secret agent

$D(y, x)$: y is the nemesis of x

We can express the statement "some secret agent has a nemesis" as:

$$\exists x (P(x) \rightarrow \exists y D(y, x))$$

this is equivalent to

$$\exists y \exists x (P(x) \rightarrow D(y, x))$$

which means "someone is the nemesis of some secret agent"

I've written the English a little differently, but they are logically the same.

Moving Quantifiers

You may move a quantifier through a logical statement as long as you don't cross another quantifier or a variable of the same name.

We cannot move across variables of the same name:

For example there is something that makes $P(x)$ true and there is something that makes $Q(x)$ true can be written as:

$$\exists x P(x) \wedge \exists x Q(x)$$

this is NOT the same as:

$$\exists x \exists x (P(x) \wedge Q(x))$$

which is a poorly written way of saying there is something that makes both $P(x)$ and $Q(x)$ true.

Prenex Normal Form (PNF)

A quantified statement is in Prenex Normal Form (PNF) if all the quantifiers appear first.

For example these are in PNF:

$$\exists x \forall y (P(x) \wedge Q(y))$$

$$\exists x \exists y \forall z ((P(x, y) \vee (Q(z)) \rightarrow R(x, y, z))$$

These are not in PNF:

$$\exists x (P(x) \rightarrow \exists y D(y, x))$$

$$\exists x P(x) \wedge \exists x Q(x)$$

For every quantified statement, there is an equivalent quantified statement that is in PNF.

Casually: “Every quantified statement can be written in PNF.”

Prenex Normal Form (PNF)

For every quantified statement, there is an equivalent quantified statement that is in PNF.

Casually: “Every quantified statement can be written in PNF.”

How?

If every quantifier has a different variable, then just push them all to the left without crossing any of the quantifiers.

$$\forall x (P(x) \rightarrow \exists y D(y, x))$$

$$\forall x \exists y (P(x) \rightarrow D(y, x))$$

Prenex Normal Form (PNF)

For every quantified statement, there is an equivalent quantified statement that is in PNF.

Casually: “Every quantified statement can be written in PNF.”

What do we do if some names are the same?

We first change the variables names so that no quantifier has the same variable name

$$\exists x P(x) \wedge \exists x Q(x)$$

$$\exists x P(x) \wedge \exists y Q(y)$$

$$\exists x \exists y (P(x) \wedge Q(y))$$

Prenex Normal Form (PNF)

There is a kind of weird case with implication and quantifiers.

Consider: $\exists x P(x) \rightarrow \exists y Q(y) \equiv (\exists x P(x)) \rightarrow \exists y Q(y)$

which means "if there is an x which makes $P(x)$ true, then there is a y which makes $Q(y)$ true"

This is not the same as $\exists x \exists y (P(x) \rightarrow \exists y Q(y))$

which means "there is an x and a y such that $P(x)$ implies $Q(y)$ ".

Notice this will be true if any x exists that makes $P(x)$ false.

So what's going on, and how do we write $\exists x P(x) \rightarrow \exists y Q(y)$ in PNF?

Prenex Normal Form (PNF)

$$\exists x P(x) \rightarrow \exists y Q(y) \equiv \neg(\exists x P(x)) \vee \exists y Q(y)$$

$$\equiv \forall x (\neg P(x)) \vee \exists y Q(y)$$

$$\equiv \forall x \exists y (\neg P(x) \vee Q(y))$$

$$\equiv \forall x \exists y (P(x) \rightarrow Q(y))$$

which means for every x , there exists a y , so that if $P(x)$ is true, then $Q(y)$ is true.

So, if a quantifier is bound to the hypothesis of a implication, and not to the entire implication, it is actually behind a negation.

Thinking of Quantifiers as Loops

The book offers this tip which might help understand quantifiers. Given the same situation as before:

$P(x)$: x is a secret agent

$D(y, x)$: y is the nemesis of x

$\exists x \forall y (P(x) \rightarrow D(y, x))$

If the domains are finite, you can imagine a program that tests the truth of the statement by using nested loops:

The above would be

```
for each x in D1:
    for each y in D2:
        if not P(x) and D(y, x) then:
            break //try next x
    return true //all y true for some x
return false
```

Thinking of Quantifiers as Loops

How would you think of

$\forall x \forall y (P(x, y))$?

```
for each x in D1:
    for each y in D2:
        if not P(x,y) then:
            return false    //found some false pair
return true
```

Thinking of Quantifiers as Loops

How would you think of

$\exists x \exists y (P(x, y))$?

```
for each x in D1:
    for each y in D2:
        if P(x,y) then:
            return true      //found some true pair
return false
```

Thinking of Quantifiers as Loops

How would you think of

$\forall x \exists y (P(x, y))$?

```
for each x in D1:
    for each y in D2:
        if P(x,y) then:
            break          //found y for this x, try the next x
    return false          //no y for this x
return true
```

End of Section 1.5
Starting Section 1.6

Rules of Inference

- An *argument* is a sequence of statements (propositions) called the *hypotheses* (or *premises*), followed by a statement (proposition) called the *conclusion*.
- An argument is called *valid* if all the propositions being true implies the conclusion must be true.

Valid Argument

p_1

p_2

...

p_n

$\therefore C$

<- read "therefore c"

$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow c$ is a tautology

Valid Argument

p_1

p_2

...

p_n

$\therefore C$

We can show an argument is valid via a truth table:

We just need to make sure c is true on all rows

where $p_1, p_2, \dots, p_n$ are all true

Valid Argument

We can show this argument is valid using a truth table:

Example:

$p \rightarrow q$

$q \rightarrow r$

$\therefore p \rightarrow r$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Valid Argument

We only care about the rows where everything in the hypothesis is true

Example:

$p \rightarrow q$

$q \rightarrow r$

$\therefore p \rightarrow r$



p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Valid Argument

If the conclusion is true in each of those rows, the argument is valid

Example:

$p \rightarrow q$

$q \rightarrow r$

$\therefore p \rightarrow r$



p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Valid Argument

To show an argument is not valid, you just need to show one case where all propositions in the hypothesis are true, but the conclusion is not.

Example:

$p \rightarrow r$
 $q \rightarrow r$

 $\therefore p \rightarrow q$

Let:
 $p = T$
 $q = F$
 $r = T$

Then:
 $p \rightarrow r$ is T
 $q \rightarrow r$ is T
but $p \rightarrow q$ is F

So the
argument is not
valid!

Classic Philosophy Example

- All Cats have four legs
- Mindy is a cat.
- Therefore, Mindy has four legs.

Classical, valid argument in English

Another Example

- f: You can fly
- s: You are Superman

Argument:

$s \rightarrow f$

s

$\therefore f$

Is it valid?

If you are Superman, then you can fly.
You are Superman. Therefore, you can fly.

Another Example

- f: You can fly
- s: You are Superman

Argument:

$s \rightarrow f$

s

$\therefore f$

Is it valid?

yes because

$(s \wedge (s \rightarrow f)) \rightarrow f$ is a tautology

If you are Superman, then you can fly.
You are Superman. Therefore, you can fly.

Valid vs Sound

- We've seen the definition for a valid arguments.
 - p, q, r, s, c is *valid* iff $p \wedge q \wedge r \wedge s \rightarrow c$ is a tautology
- An argument is *sound* iff it is valid and all the premises are true
- For example: If I am Superman then I can fly. I can fly. Therefore I am Superman. Is NOT valid (or sound)
- If I have a fedora, then I am Perry the Platypus. I have a fedora. Therefore, I am Perry the Platypus. Is valid, but not sound.

Rules of Inference

- Some established, relatively simple valid argument forms are gathered together and called the *rules of inference*.

Rules of Inference

Name	premises	conclusion
Modus ponens	$p, p \rightarrow q$	q
Modus tollens	$\neg q, p \rightarrow q$	$\neg p$
Hypothetical syllogism	$p \rightarrow q, q \rightarrow r$	$p \rightarrow r$
Disjunctive syllogism	$p \vee q, \neg p$	q
Addition	p	$p \vee q$
Simplification	$p \wedge q$	p
Conjunction	p, q	$p \wedge q$
Resolution	$p \vee q, \neg p \vee r$	$q \vee r$

Rules of Inference Examples

- If I am superman, then I have x-ray vision. If I have x-ray vision I can see who's outside. Therefore, if I am Superman, I can see who's outside

s: I am Superman

x: I have x-ray vision

o: I can see who is outside

$s \rightarrow x$

$x \rightarrow o$

$\therefore s \rightarrow o$

Rules of Inference Examples

- If I am superman, then I have x-ray vision. If I have x-ray vision I can see who's outside. Therefore, if I am Superman, I can see who's outside

s: I am Superman

x: I have x-ray vision

o: I can see who is outside

$s \rightarrow x$

$x \rightarrow o$

$\therefore s \rightarrow o$

Hypothetical syllogism

Rules of Inference Examples

- If I am superman, then I'm faster than a speeding bullet. I am Superman.
Therefore, I am faster than a speeding bullet.

s: I am Superman

f: I am faster than a speeding bullet

$s \rightarrow f$

s

$\therefore f$

Rules of Inference Examples

- If I am superman, then I'm faster than a speeding bullet. I am Superman.
Therefore, I am faster than a speeding bullet.

s: I am Superman

f: I am faster than a speeding bullet

$s \rightarrow f$

s

$\therefore f$

Modus ponens

Rules of Inference Examples

- Either Superman will defeat Batman or Batman will have kryptonite. Batman defeats Superman. Therefore, Batman has kryptonite.

s: Superman defeats Batman

k: Batman has kryptonite

$s \vee k$

$\neg s$

$\therefore k$

Rules of Inference Examples

- Either Superman will defeat Batman or Batman will have kryptonite. Batman defeats Superman. Therefore, Batman has kryptonite.

s: Superman defeats Batman

k: Batman has kryptonite

$s \vee k$

$\neg s$

$\therefore k$

Disjunctive syllogism

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.
- Can we conclude that, if I don't send you flowers, I'll do well on the quiz?

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.
- Can we conclude that, if I don't send you flowers, I'll do well on the quiz?

Begin by identifying the propositions:

p: You email me a love note

q: I send you flowers

r: I study Discrete Math

s: I do well on the quiz

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.
- Can we conclude that, if I don't send you flowers, I'll do well on the quiz?

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

p: You email me a love note

q: I send you flowers

r: I study Discrete Math

s: I do well on the quiz

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

1. $p \rightarrow q$

hypothesis

$p \rightarrow q$

$\neg p \rightarrow r$

$r \rightarrow s$

Conclusion:

$\therefore \neg q \rightarrow s$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

1. $p \rightarrow q$

hypothesis

2. $\neg q \rightarrow \neg p$

contrapositive (1)

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

1. $p \rightarrow q$

hypothesis

2. $\neg q \rightarrow \neg p$

contrapositive (1)

3. $\neg p \rightarrow r$

hypothesis

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

1. $p \rightarrow q$

2. $\neg q \rightarrow \neg p$

3. $\neg p \rightarrow r$

4. $\neg q \rightarrow r$

hypothesis

contrapositive (1)

hypothesis

hypothetical syllogism (2, 3)

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

1. $p \rightarrow q$

hypothesis

2. $\neg q \rightarrow \neg p$

contrapositive (1)

3. $\neg p \rightarrow r$

hypothesis

4. $\neg q \rightarrow r$

hypothetical syllogism (2, 3)

5. $r \rightarrow s$

hypothesis

Conclusion:

$$\therefore \neg q \rightarrow s$$

Example

- If you email me a love note, I'll send you flowers. If you don't, I'll study Discrete Math. If I study Discrete Math, I'll do well on the quiz.

Hypothesis:

$$p \rightarrow q$$

$$\neg p \rightarrow r$$

$$r \rightarrow s$$

Conclusion:

$$\therefore \neg q \rightarrow s$$

1. $p \rightarrow q$

hypothesis

2. $\neg q \rightarrow \neg p$

contrapositive (1)

3. $\neg p \rightarrow r$

hypothesis

4. $\neg q \rightarrow r$

hypothetical syllogism (2, 3)

5. $r \rightarrow s$

hypothesis

6. $\neg q \rightarrow s$

hypothetical syllogism (4, 5)

This is called a logical proof

Rules of Inference with Quantifiers

- We now want to prove conclusions with quantified propositions.
- To apply the rules of inference in these cases we must remove the quantifier by plugging in a value from the domain for the variable.
- We may either plug in an **arbitrary** element of the domain that has no special properties, or a **particular** element that has some special properties.

Rules of Inference with Quantifiers

The domain is D

Universal Instantiation:

$\forall x P(x)$ $\therefore P(c)$ (for any arbitrary or particular $c \in D$)

Example:

D = all cats

Every cat has four legs.

Mindy is a cat.

Therefore, Mindy has four legs

Rules of Inference with Quantifiers

The domain is D

Universal Generalization:

c is an arbitrary element of D

P(c)

$\therefore \forall x P(x)$

Example:

D = all cats

Let c be an arbitrary cat

c has four legs

Therefore, every cat has four legs

Rules of Inference with Quantifiers

The domain is D

Existential Instantiation:

$\exists x P(x)$

$\therefore P(c)$ some particular $c \in D$

Example:

D = all cats

There is a cat with a grey nose

Mindy is a cat with a grey nose

Rules of Inference with Quantifiers

The domain is D

Existential Generalization:

$P(c)$

$\therefore \exists x P(x)$

some particular/arbitrary $c \in D$

Example:

D = all cats

Mindy is a cat with a grey nose

There is a cat with a grey nose

Rules of Inference for Predicates

Example(s):

Everyone in this class has had a programming course.

Hugo is in this class. Has he had a programming course?

Rules of Inference for Predicates

Example(s):

Everyone in this class has had a programming course.
Hugo is in this class. Has he had a programming course?

Yes! Here's how we know:

$C(x)$: x is in this class, $x \in \text{People}$

$P(x)$: x has had a programming course, $x \in \text{People}$

(1) $\forall x (C(x) \rightarrow P(x))$ (Hypothesis)

(2) $C(\text{Hugo})$ (Hypothesis)

(3) $C(\text{Hugo}) \rightarrow P(\text{Hugo})$ (1, Univ. Instantiation)

(4) $\therefore P(\text{Hugo})$ (2, 3, Modus Ponens)

Identify the error in the argument that supposedly shows that if

$\exists x P(x) \wedge \exists x Q(x)$ then $\exists x (P(x) \wedge Q(x))$

Identify the error in the argument that supposedly shows that if
 $\exists x P(x) \wedge \exists x Q(x)$ then $\exists x (P(x) \wedge Q(x))$

- | | |
|---|------------------------------------|
| 1. $\exists x P(x) \wedge \exists x Q(x)$ | Premise |
| 2. $\exists x P(x)$ | Simplification from (1) |
| 3. $P(c)$ | Existential instantiation from (2) |
| 4. $\exists x Q(x)$ | Simplification from (1) |
| 5. $Q(c)$ | Existential instantiation from (4) |
| 6. $P(c) \wedge Q(c)$ | Conjunction of (3) and (5) |
| 7. $\exists x (P(x) \wedge Q(x))$ | Existential generalization |

Identify the error in the argument that supposedly shows that if
 $\exists x P(x) \wedge \exists x Q(x)$ then $\exists x (P(x) \wedge Q(x))$

1. $\exists x P(x) \wedge \exists x Q(x)$	Premise
2. $\exists x P(x)$	Simplification from (1)
3. $P(c)$	Existential instantiation from (2)
4. $\exists x Q(x)$	Simplification from (1)
5. $Q(c)$	Existential instantiation from (4)
6. $P(c) \wedge Q(c)$	Conjunction of (3) and (5)
7. $\exists x (P(x) \wedge Q(x))$	Existential generalization

The problem is the c in 3 and in 5 are not necessarily the same element.



Combining Rules of Inference for Propositions and Quantified Statements

$$\forall x(P(x) \rightarrow Q(x))$$

$P(a)$, where a is a particular element in the domain

$$\therefore Q(a)$$

This is called *universal modus ponens*.



Combining Rules of Inference for Propositions and Quantified Statements

$$\forall x (P(x) \rightarrow Q(x))$$

$\neg Q(a)$, where a is a particular element in the domain

$$\therefore \neg P(a)$$

This is called *universal modus tollens*.

Specious Reasoning: The Bear Patrol

Homer: Ah, not a bear in sight. The Bear Patrol must be working like a charm!

Lisa: That's **specious reasoning**, Dad. [...] By your logic, I could claim that this rock keeps tigers away!

Homer: Uh-huh, and how does it work?

Lisa: It doesn't work. [...] It's just a stupid rock. [...] But I don't see any tigers around here, do you?

Homer: Lisa, I'd like to buy your rock.



From: **The Simpsons**, “Much Apu About Nothing”

(Season 7, Episode 151, Production Code 3F20)

Specious Reasoning: The Bear Patrol

Definition: Specious Reasoning

An unsupported or improperly constructed argument.
(That is, an unsound or invalid argument.)

Question: Where is the error in Homer's logic?

b: There are bears in Springfield

w: The Bear Patrol is working

First issue: What is Homer's argument?

Specious Reasoning: The Bear Patrol

Question: Where is the error in Homer's logic?

- (1) $\neg b$ (Given)
(2) $w \rightarrow \neg b$
—
(3) $\therefore w$

**Invalid
argument**

Fallacies

Three classic types:

1. Affirming the Conclusion

Example:

If you don't take the final, then you will fail the class. You failed the class. Therefore, you didn't take the final.

$$p \rightarrow q, q \mid \therefore p$$

Fallacies

Three classic types:

2. Denying the Hypothesis

Example:

If Margret is in Dallas, then she is in Texas. Margret isn't in Dallas. Therefore, she is not in Texas.

$$p \rightarrow q, \neg p \mid \therefore \neg q$$

Fallacies

Three classic types:

3. Begging the Question (a.k.a. Circular Reasoning)

Examples:

I'm not lying because I'm telling the truth.
It's like that because that's the way it is.

$\dots, p, \dots \mid \therefore p$

This is actually valid.
It's just not a good
"proof".